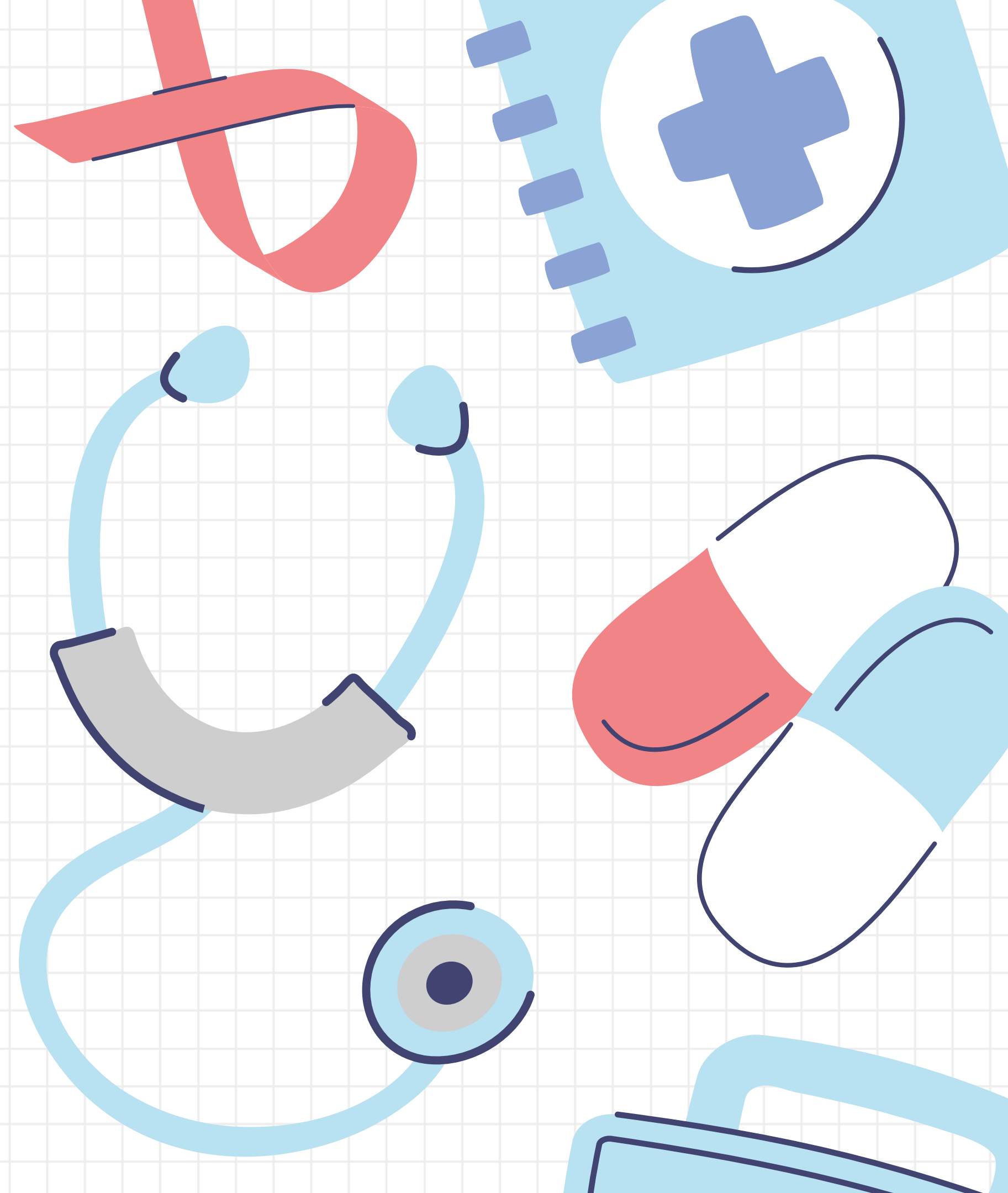
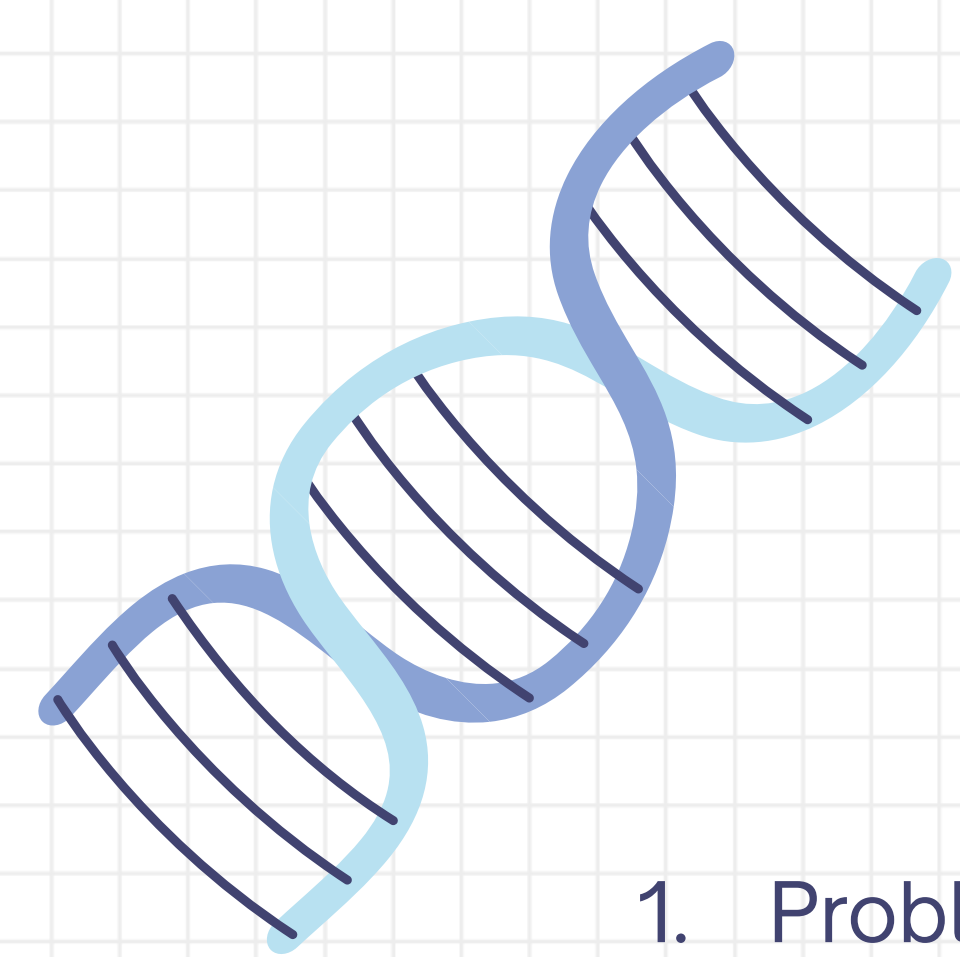


Chin Jia Li

Identifying Skin Cancer





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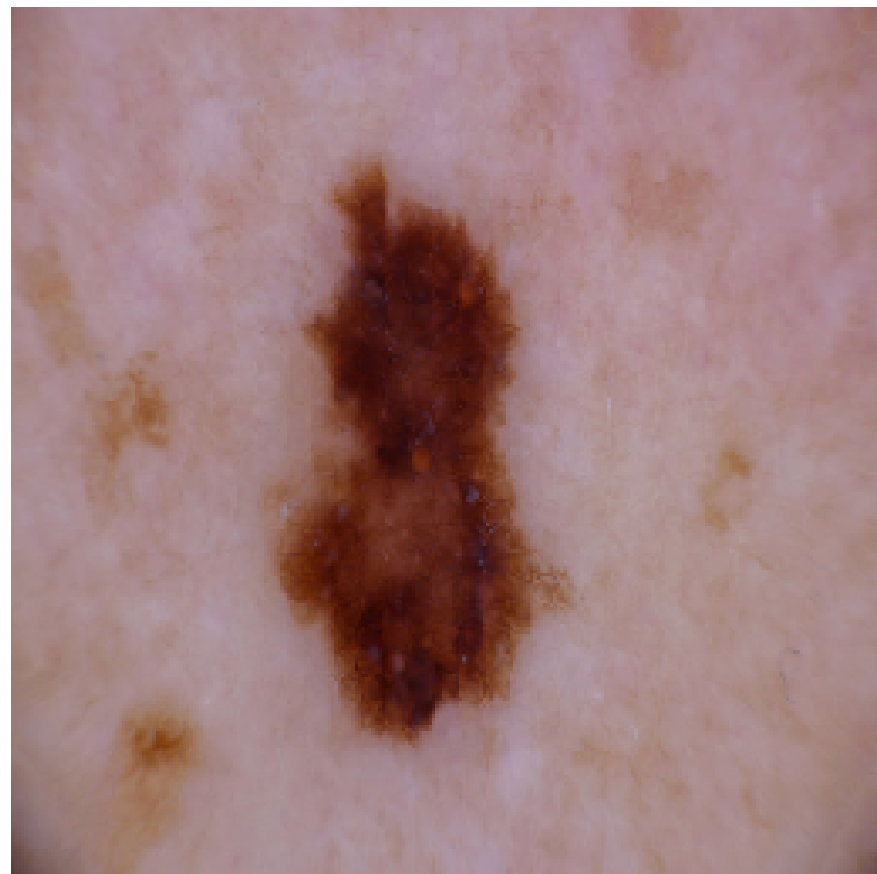


Introduction

A skin lesion is any abnormal area of skin that looks or feels different from the surrounding skin

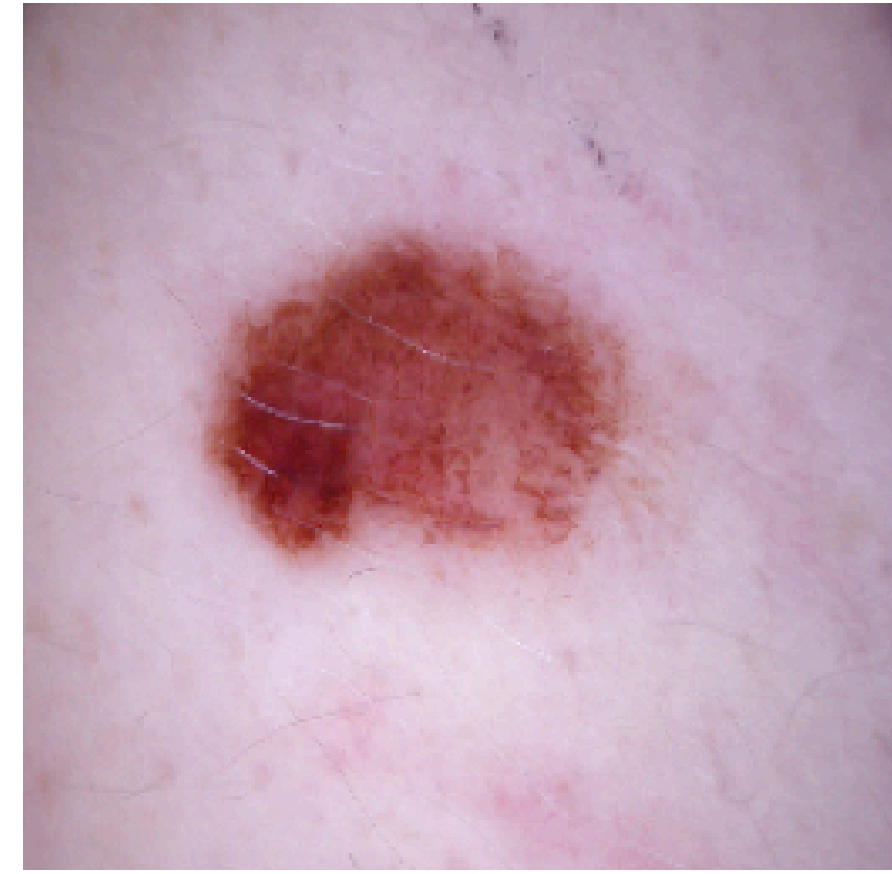
Malignant (Cancerous)

Melanoma



Benign (Non-Cancerous)

Melanocytic nevi



Problem Statement

- Difficult to differentiate between benign vs. malignant skin lesions.
- Everyone is affected by this problem, especially those that have high exposures to the sun.
- People can become more aware of skin cancer symptoms with this program and get diagnosed properly by a doctor.



🔍 Data Overview

Dataset: <https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000>

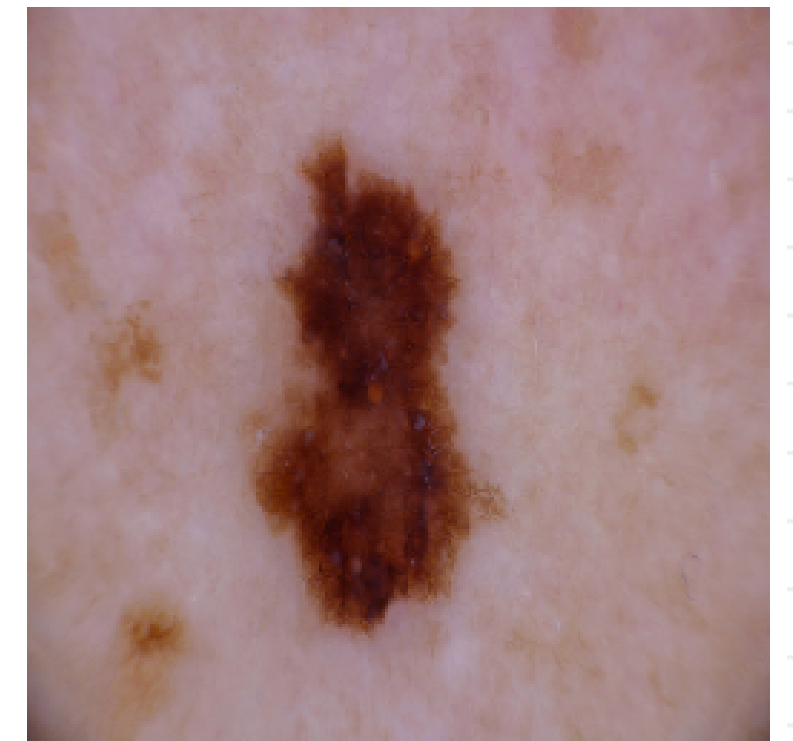
Size: 10015 images

Metadata:

	lesion_id	image_id	dx	dx_type	age	sex	localization	cell_type	image_path
0	HAM_0000118	ISIC_0027419	bkl	histo	80.0	male	scalp	Benign keratosis-like lesions	ISIC_0027419.jpg
1	HAM_0000118	ISIC_0025030	bkl	histo	80.0	male	scalp	Benign keratosis-like lesions	ISIC_0025030.jpg
2	HAM_0002730	ISIC_0026769	bkl	histo	80.0	male	scalp	Benign keratosis-like lesions	ISIC_0026769.jpg
3	HAM_0002730	ISIC_0025661	bkl	histo	80.0	male	scalp	Benign keratosis-like lesions	ISIC_0025661.jpg
4	HAM_0001466	ISIC_0031633	bkl	histo	75.0	male	ear	Benign keratosis-like lesions	ISIC_0031633.jpg

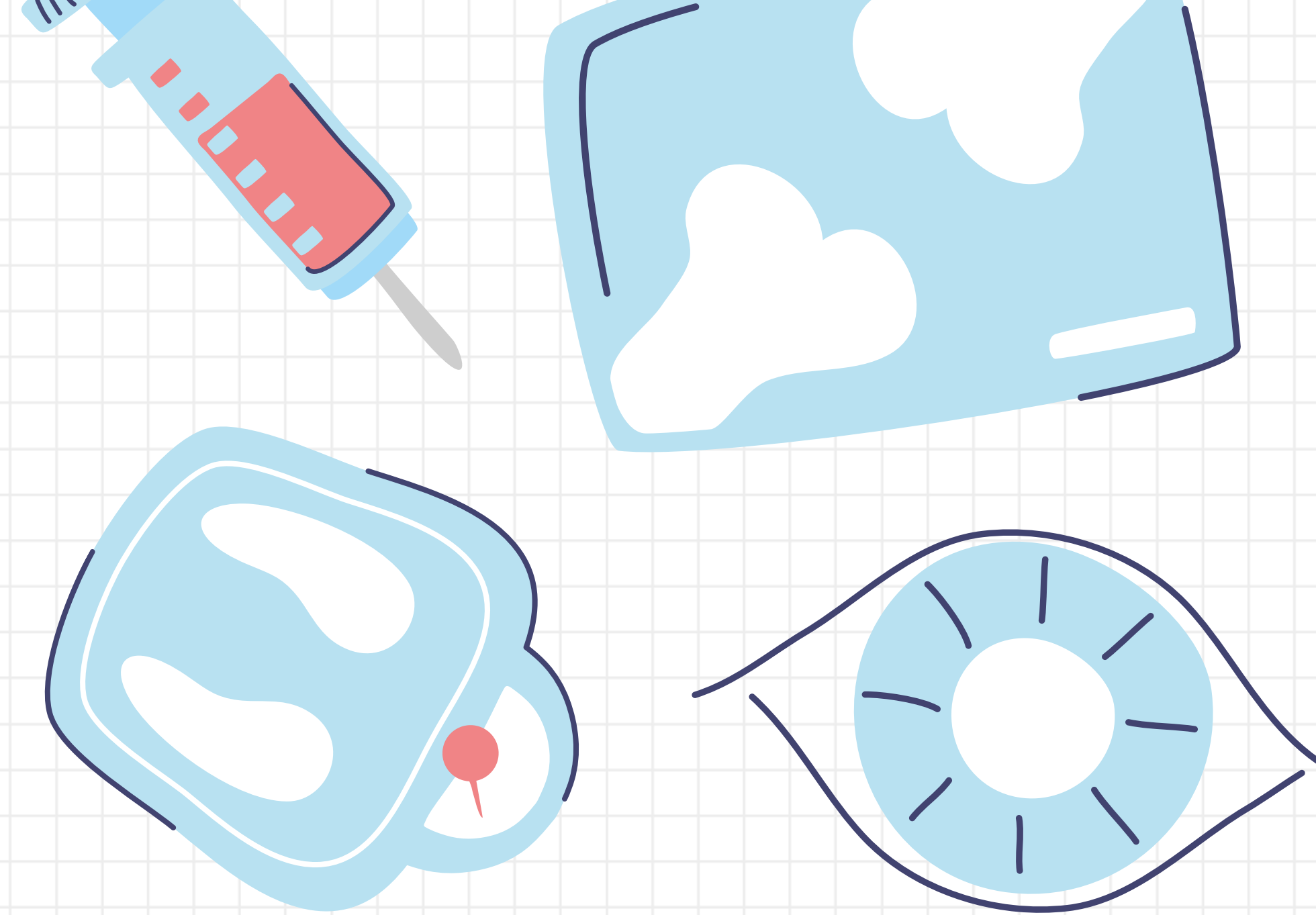
Example Image:

Melanoma



Project Objectives

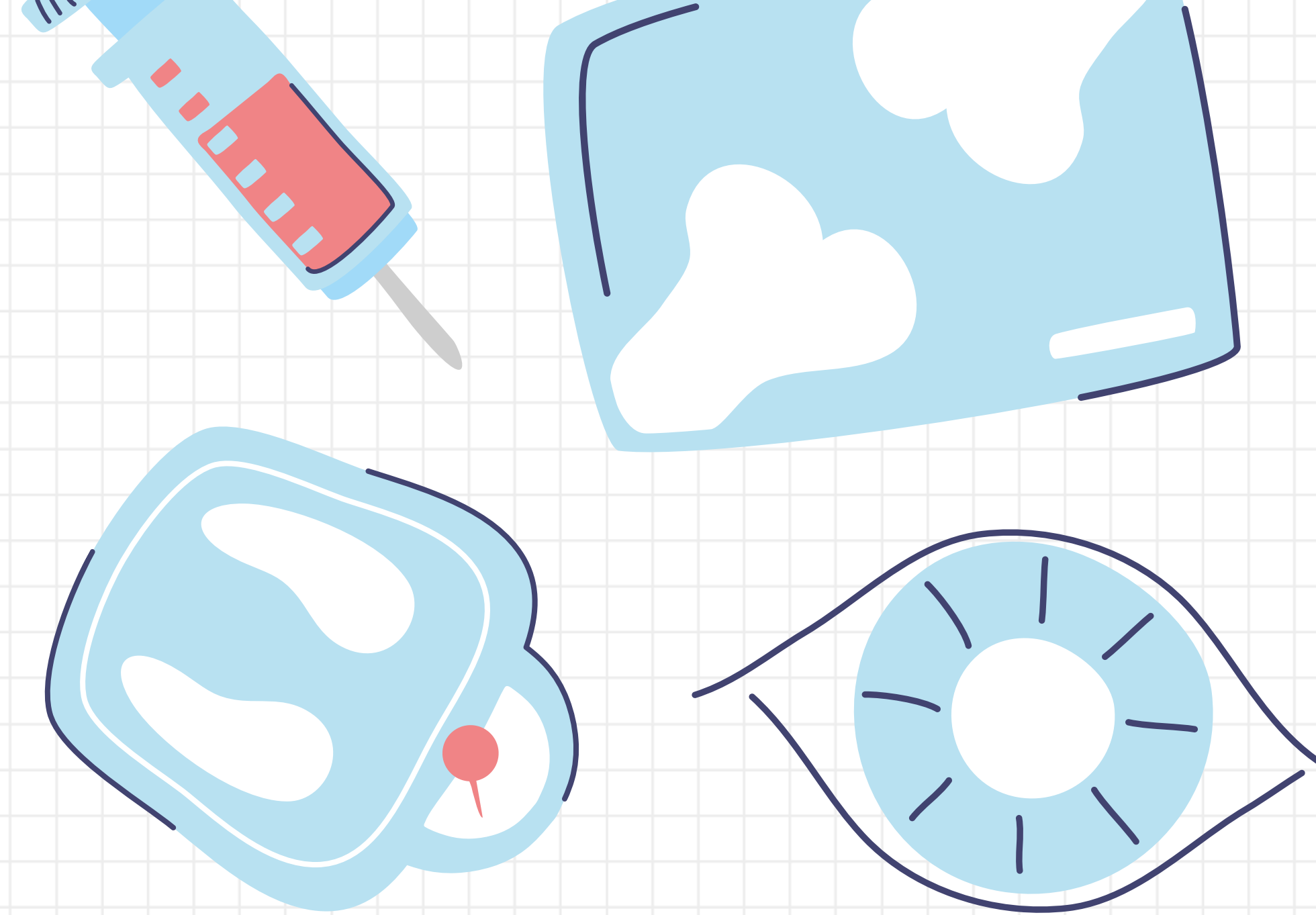
- Develop a model capable of accurately classifying different types of skin cancer with images.
- Evaluate the effectiveness of deep learning models in distinguishing between multiple skin lesion categories.



Objective & Key Questions

Key Questions

- How accurately can a deep learning model classify the seven different skin lesion types in the HAM10000 dataset?
- Which skin cancer classes are most frequently misclassified, and what does this indicate about class imbalance, visual similarity between lesions, or model limitations?



Objective & Key Questions

Methodology

Preprocessing:

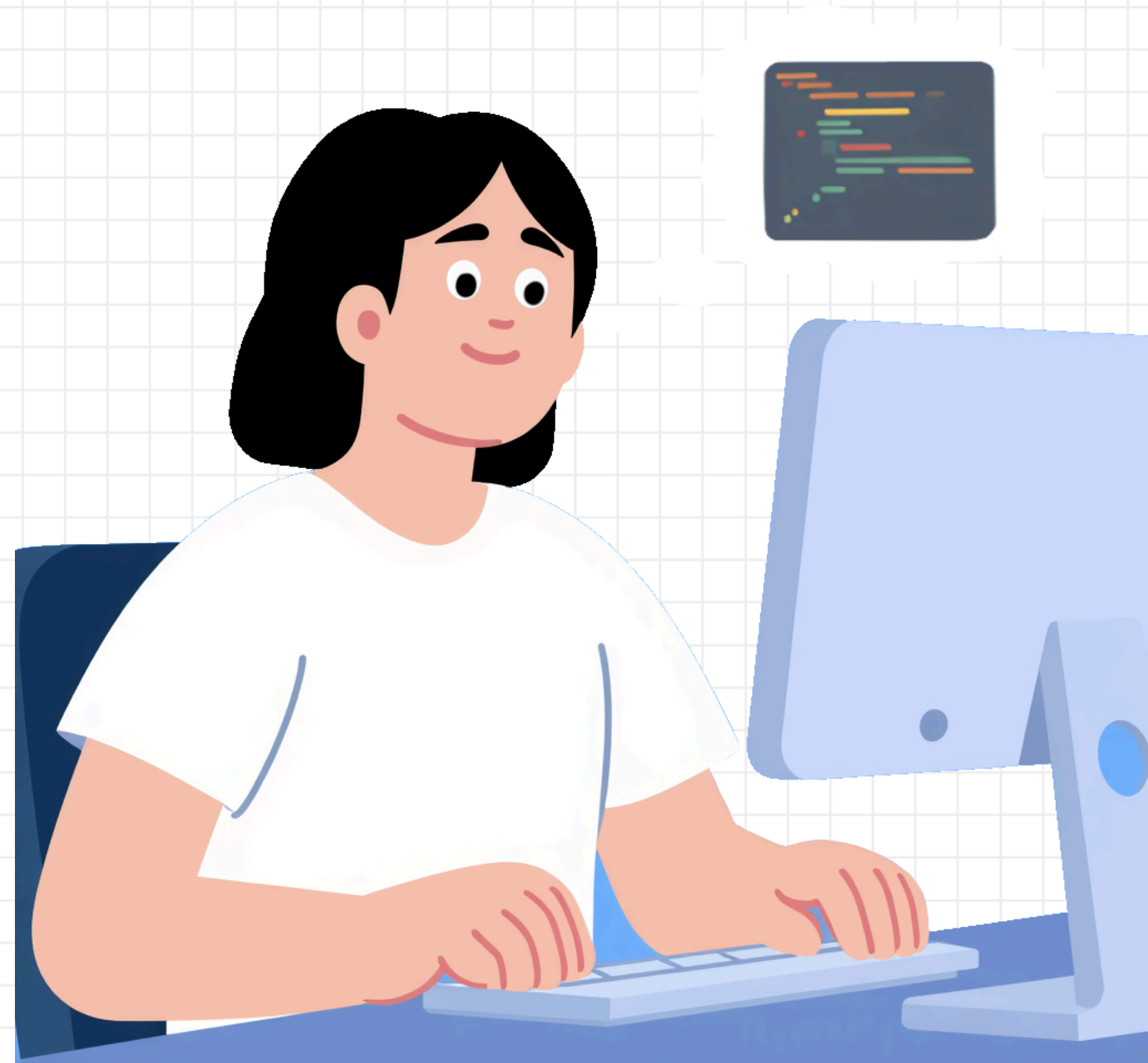
- Removing invalid values.
- Filling missing values.

Model Training:

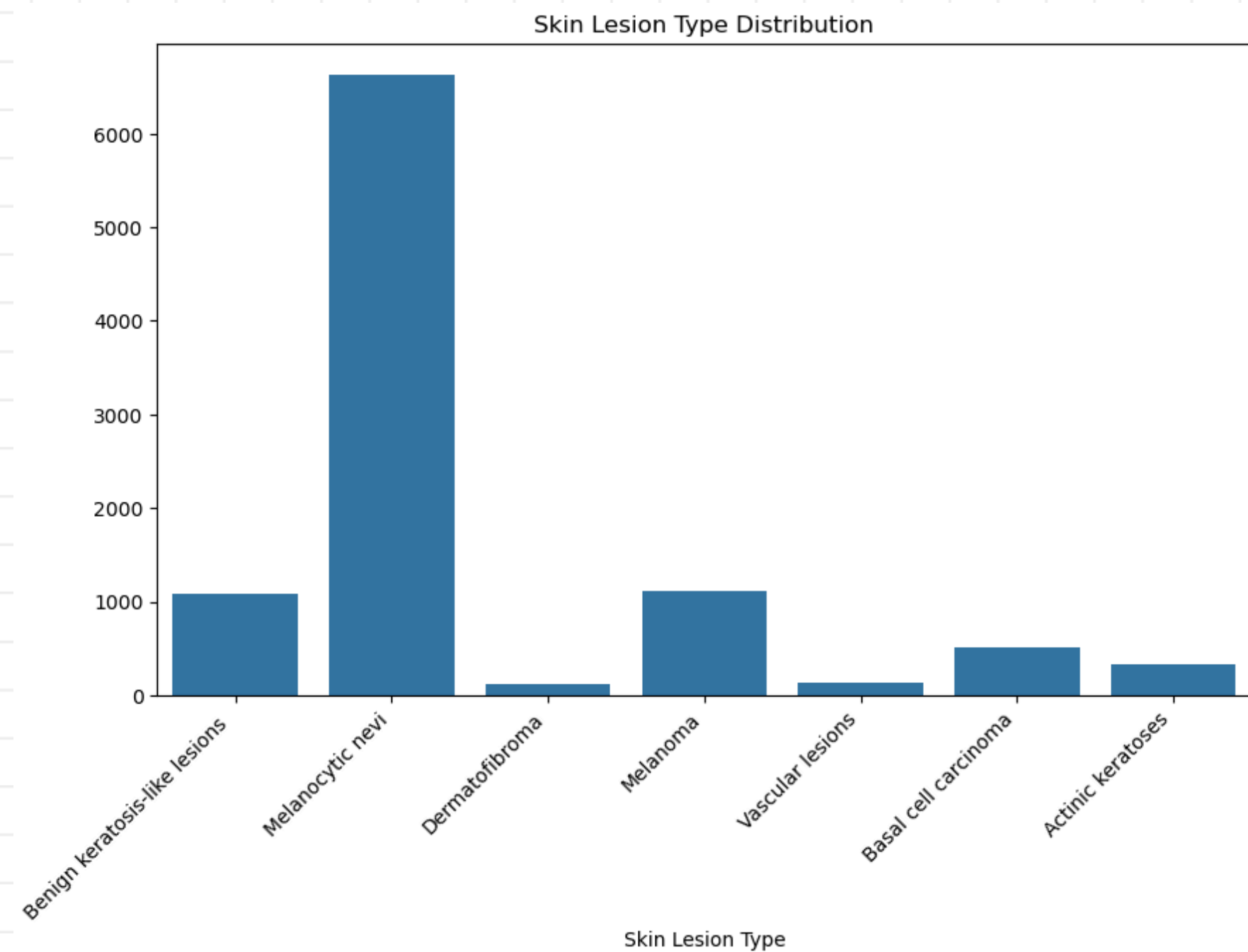
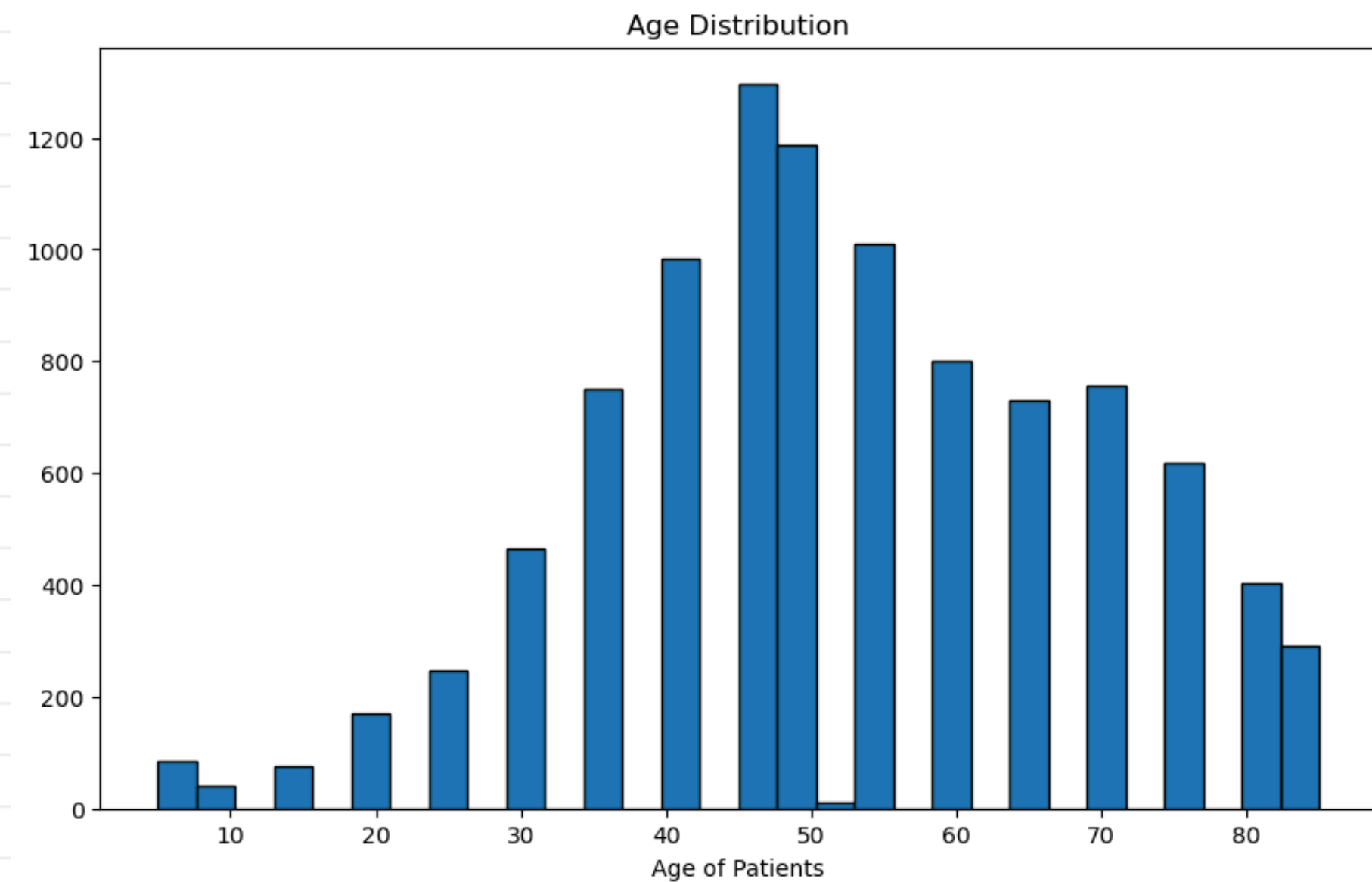
- Data generators with augmentation.
- Defined the model's layers, then compiled and fitted the model.

Evaluating:

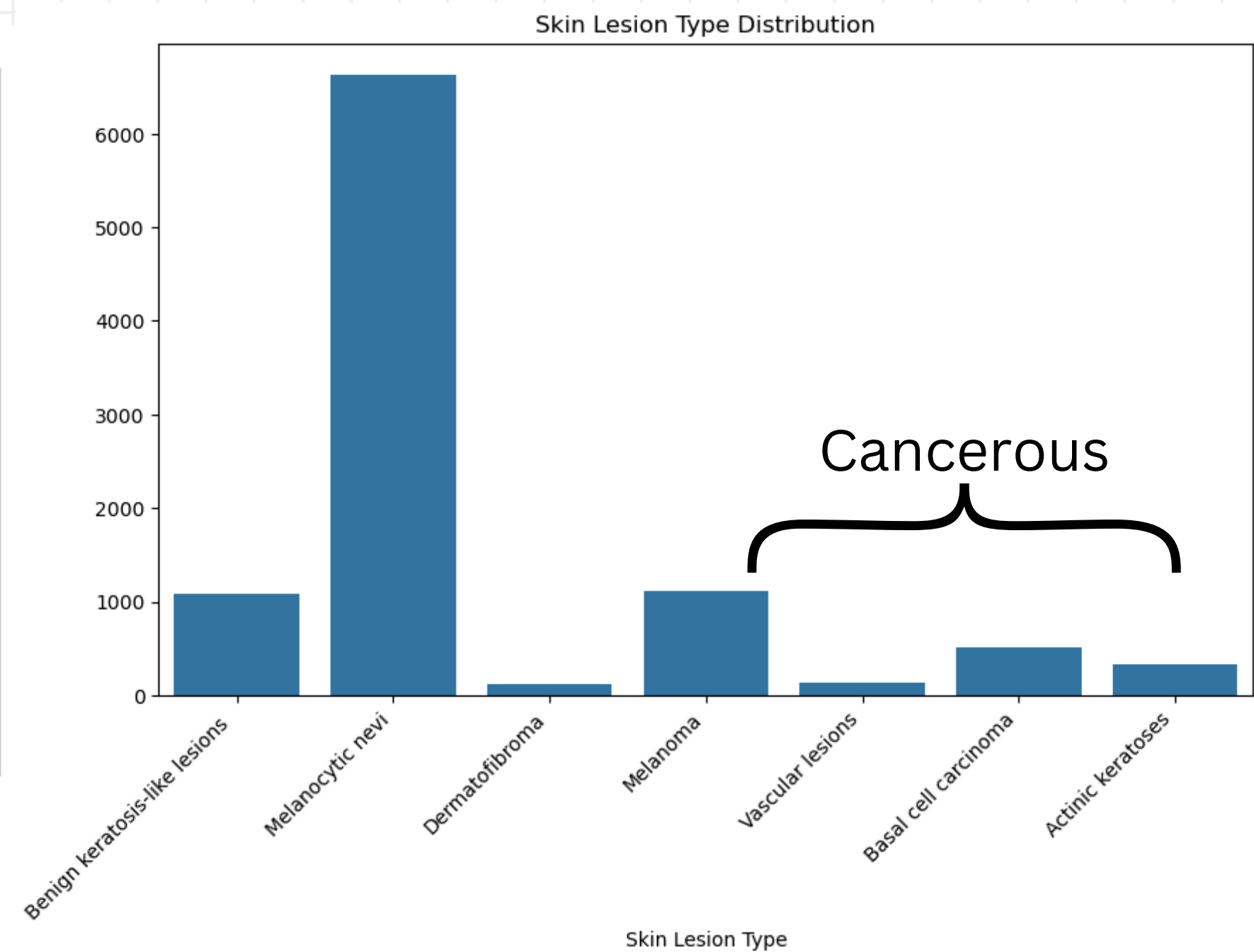
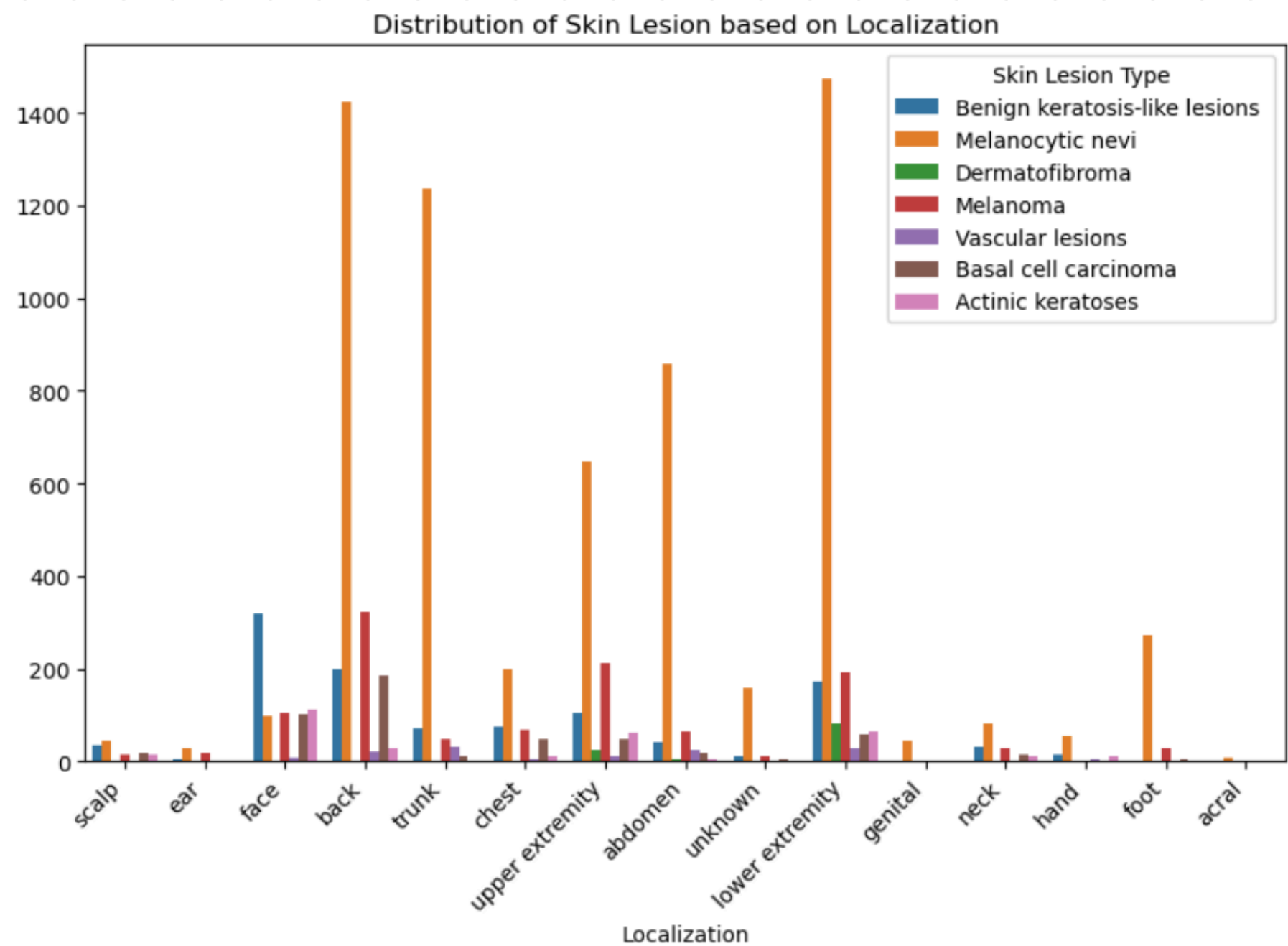
- Training loss vs Validation loss graph
- Confusion Matrix Display
- Accuracy score



EDA & Key Findings



EDA & Key Findings

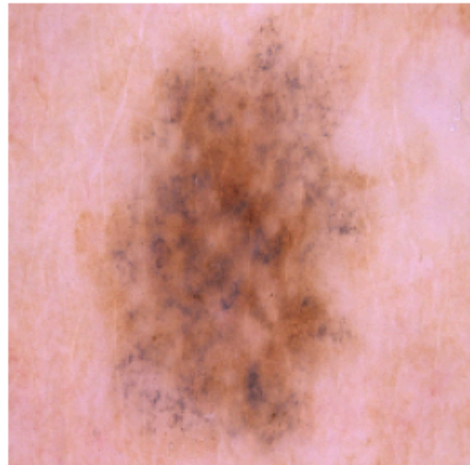


EDA & Key Findings

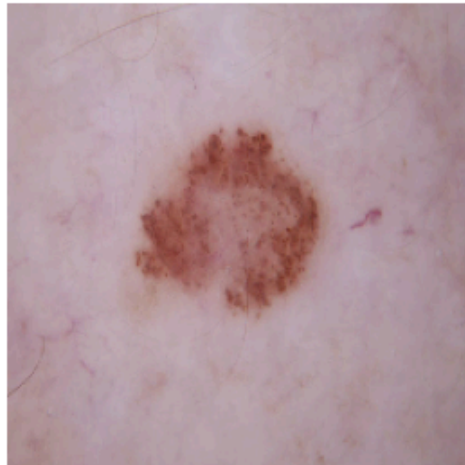
Sample images

Sample Skin Cancer Images

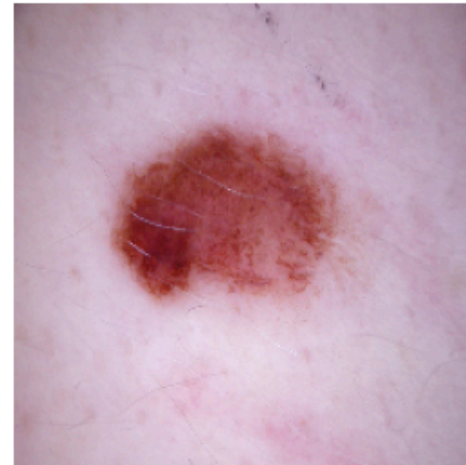
Melanocytic nevi



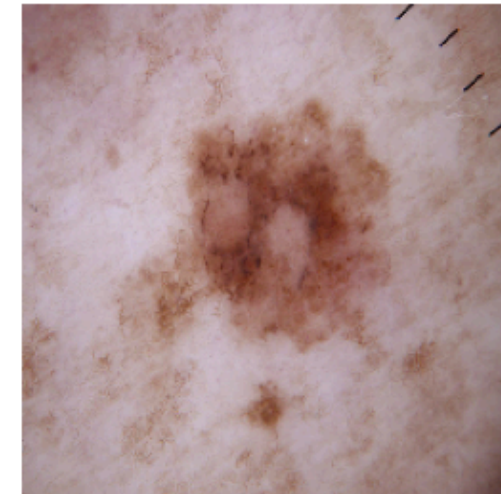
Melanocytic nevi



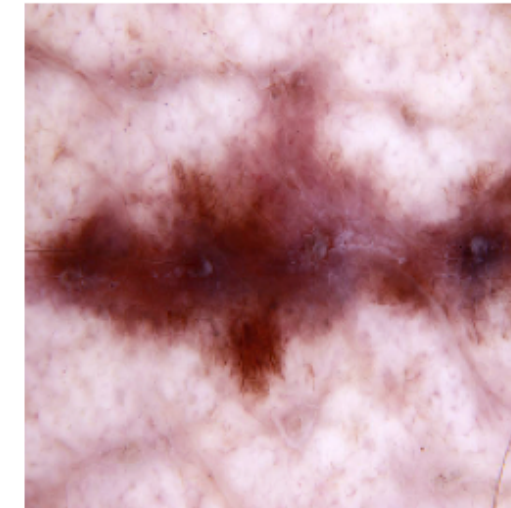
Melanocytic nevi



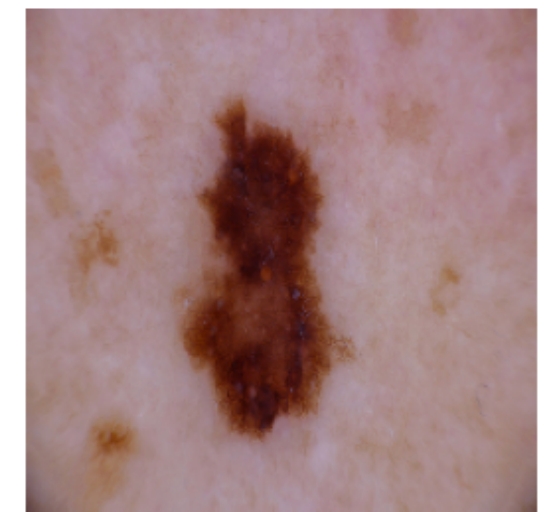
Benign keratosis-like lesions



Melanoma

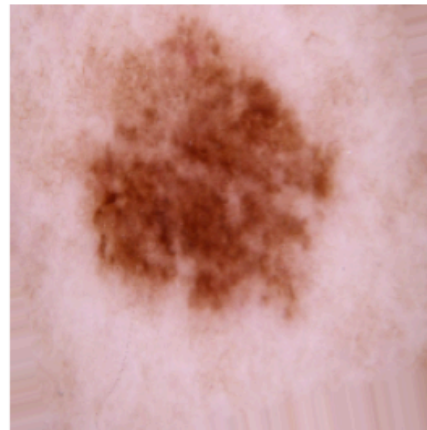


Melanoma

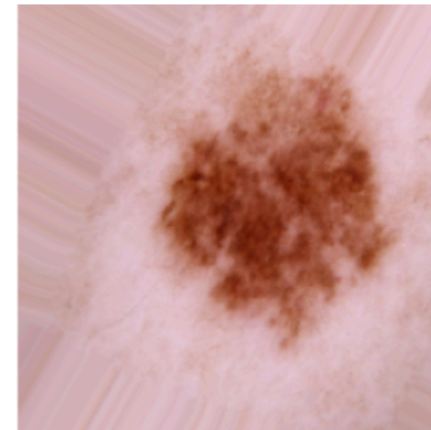


Augmented Images

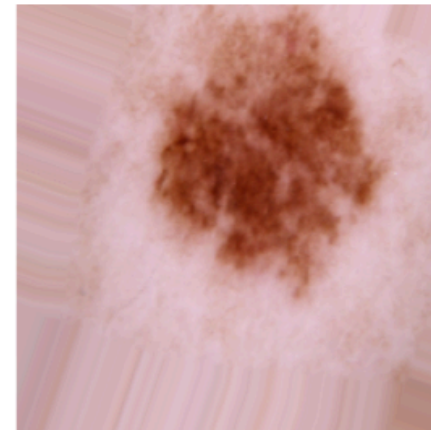
Original



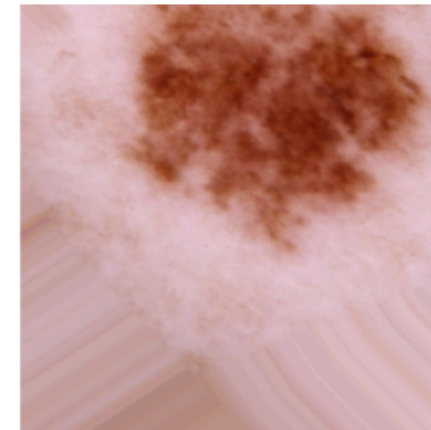
Augmented 1



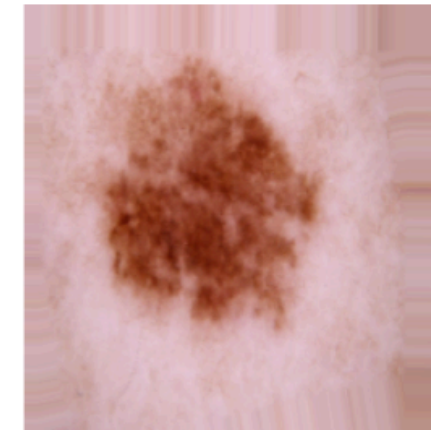
Augmented 2



Augmented 3

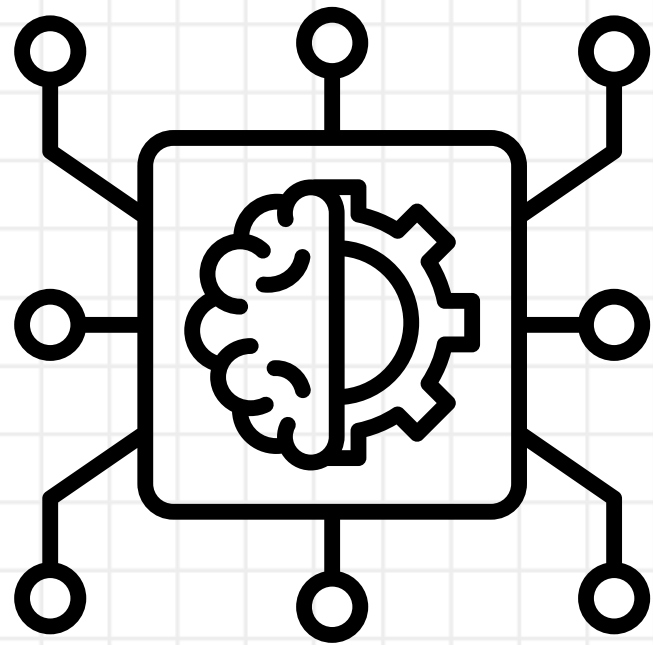


Augmented 4

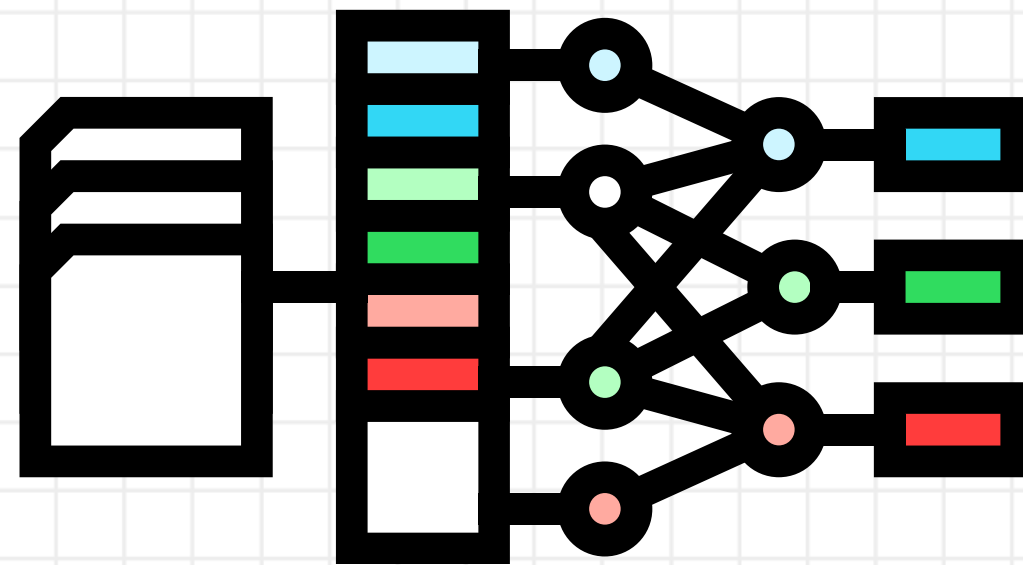


Modelling Approach

I used a 80/20 split by separating 8000 images in the training folder and 2015 images in the testing folder. I encoded the different types of skin lesion types to



Simple
CNN



Deeper
CNN

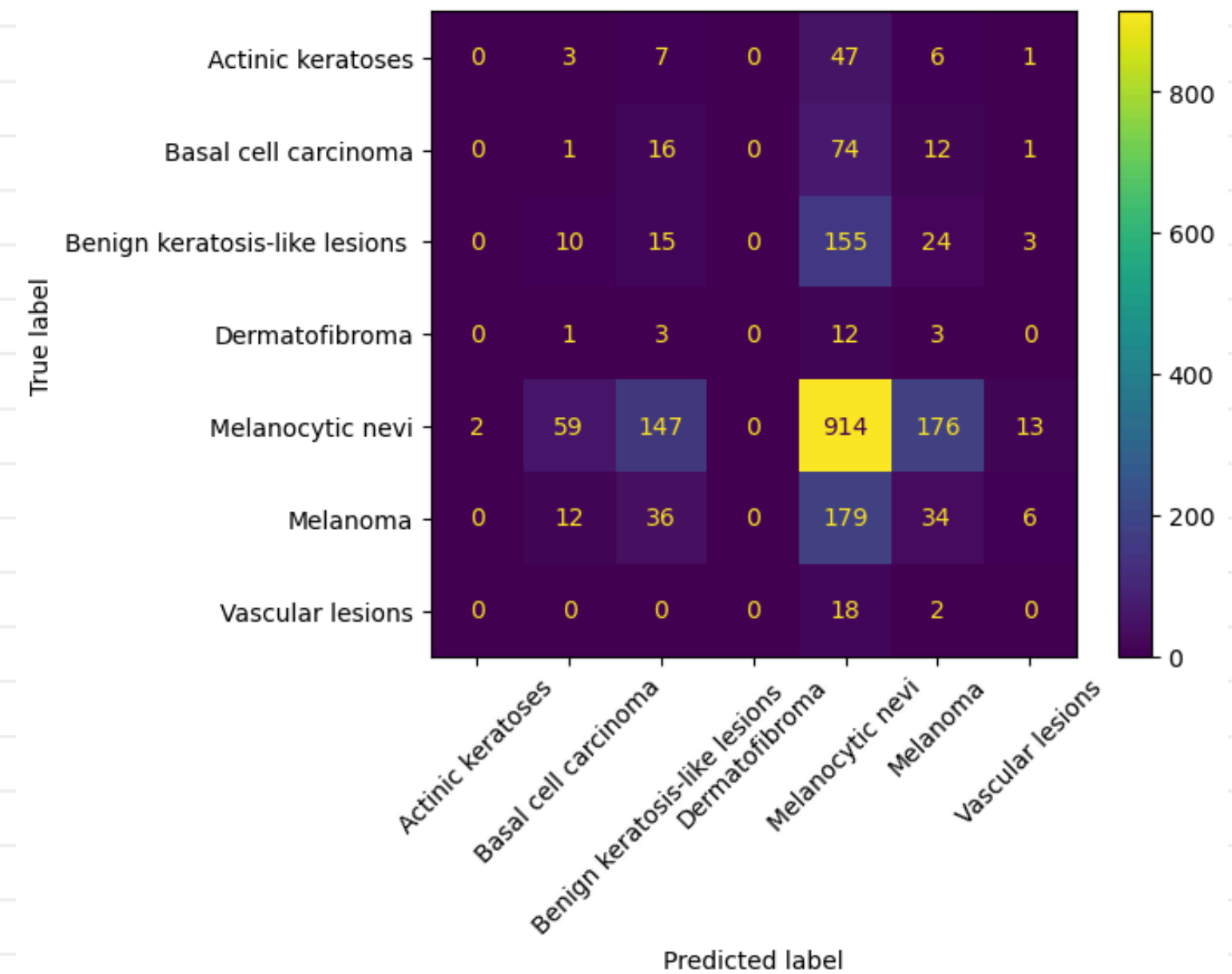
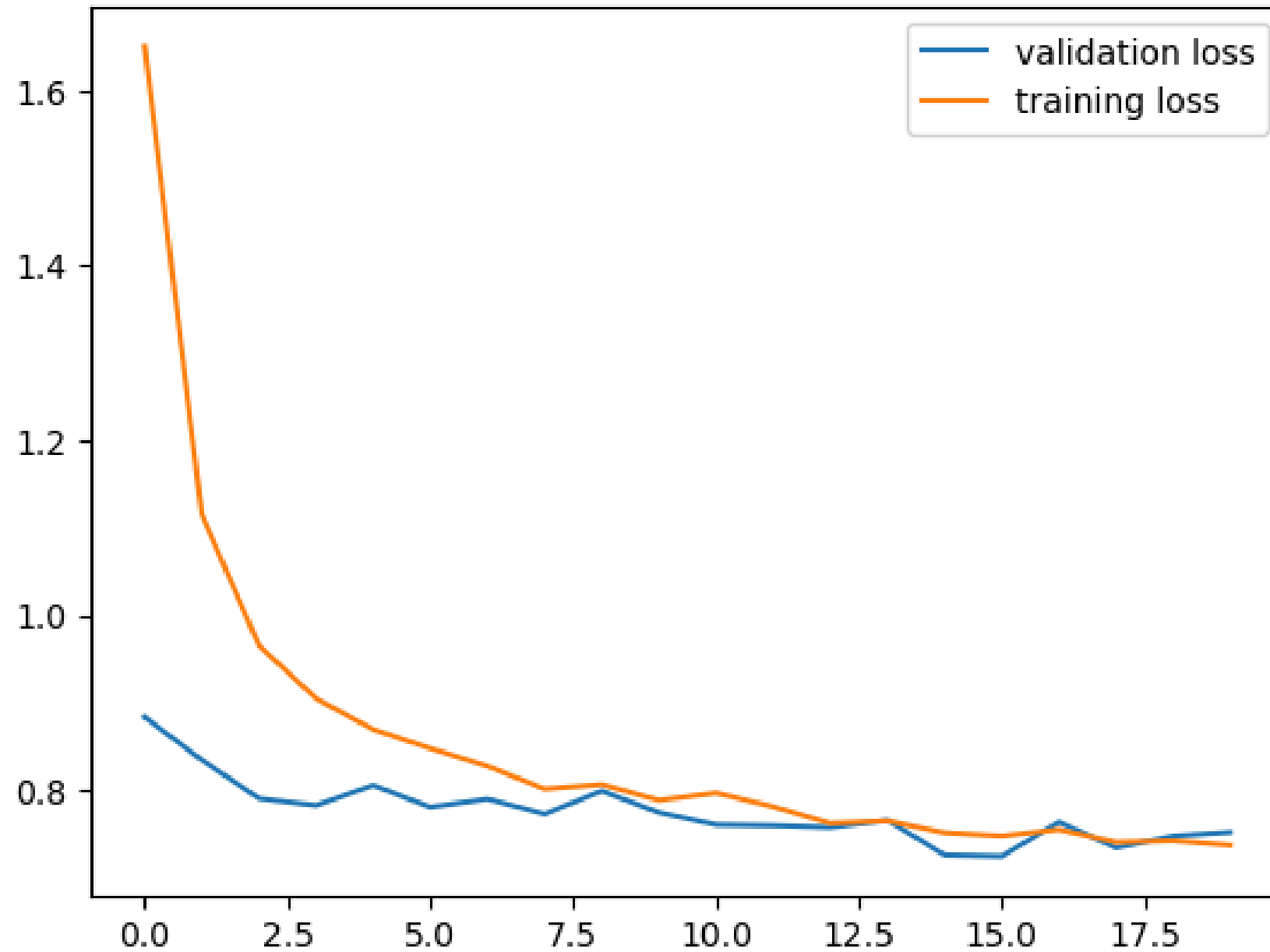


MobileNet v2

Transfer
Learning Model

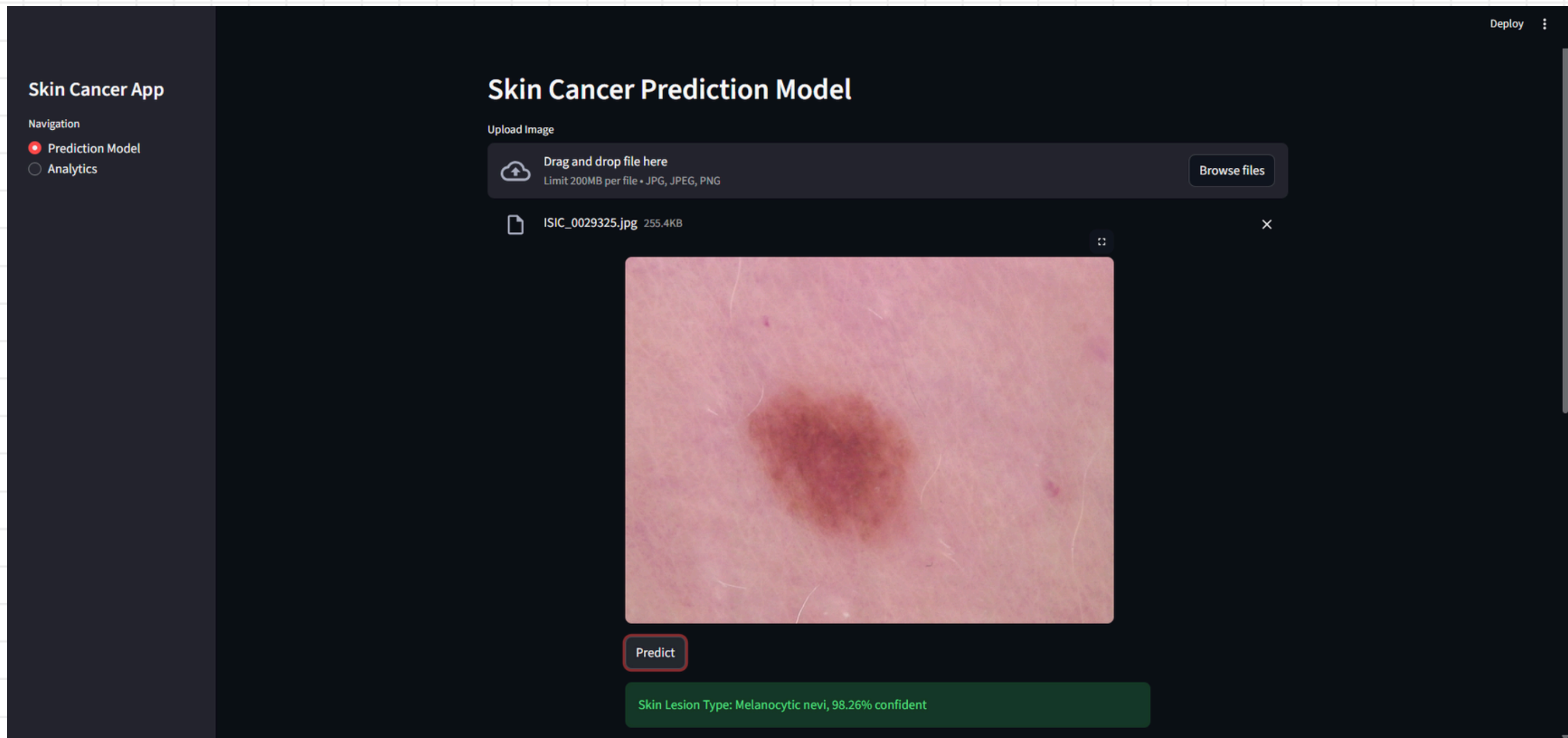
Results & Evaluation

The transfer learning model managed to achieve an accuracy score of 74.75%



This means that the model has a 74% chance of correctly identifying the type of skin lesion from the image.

Project Demo



Measure of Success

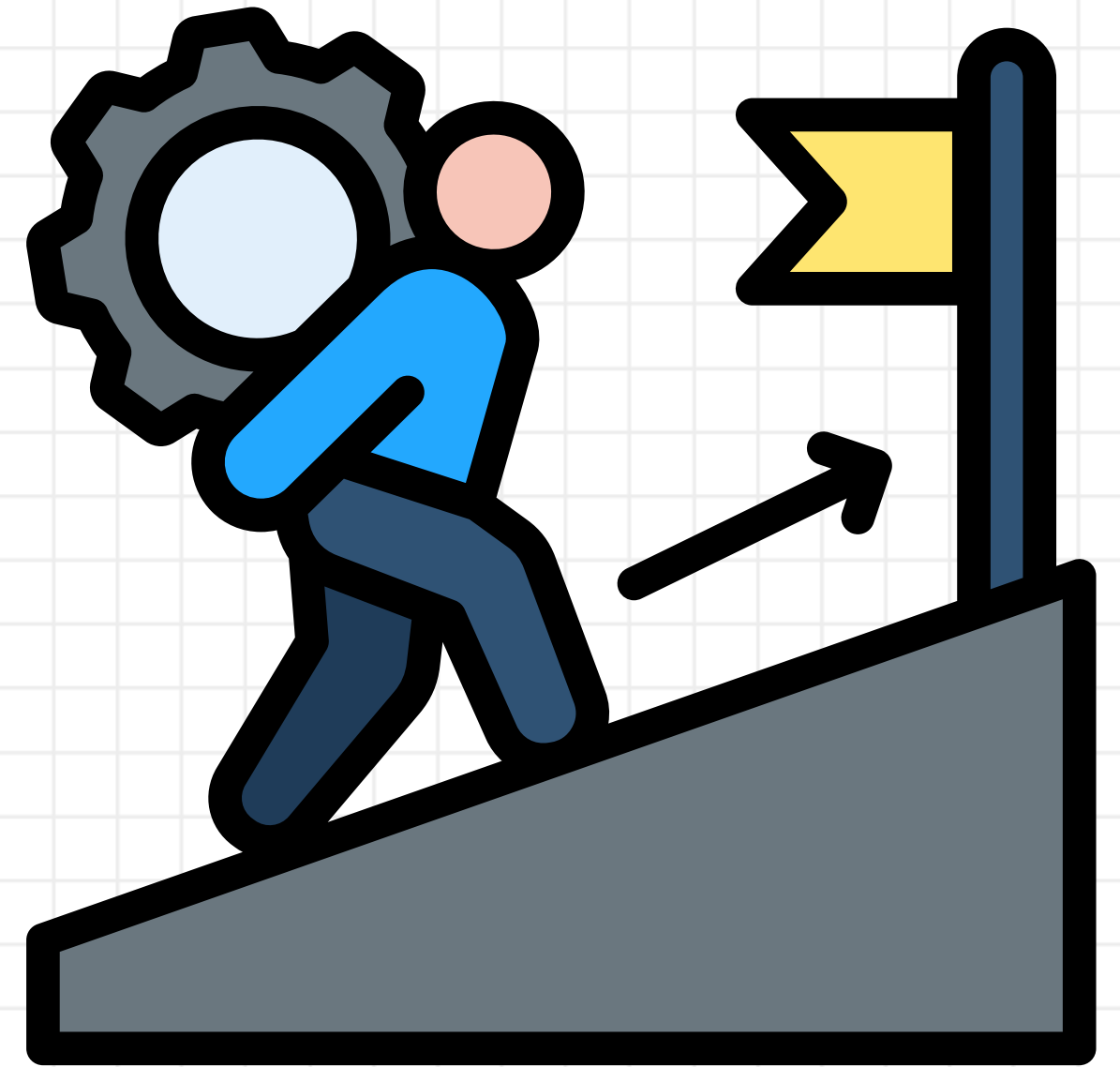
- The model is considered successful if its accuracy is more than 70%.
- The transfer learning model achieved the best accuracy score (74.75%) out of all 3 models.
- Overall, all models achieved an accuracy score of 70% and above.

Business / Clinical KPI Success Criteria:

The model demonstrates high recall for malignant skin lesions, supporting earlier identification of potentially life-threatening skin cancers.



1. Extremely skewed dataset.
2. Limited dataset diversity.
3. Image quality variation.

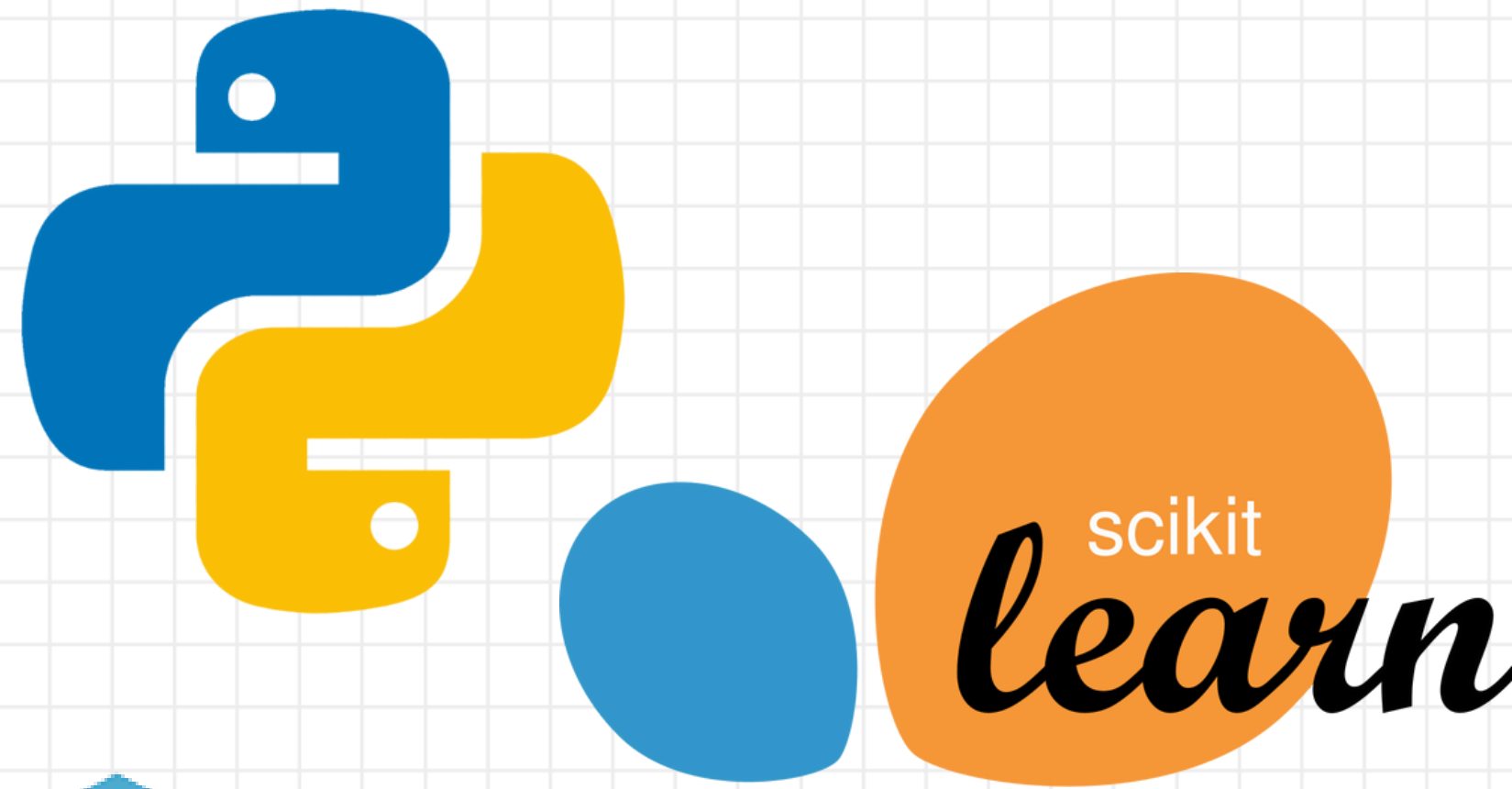


Challenges and Limitations

Future Work and Recommendations

- Run more epochs to improve accuracy
- More image augmentation settings
- Less skewed data
- Hyperparameter tuning





Tech Stack

- Language: Python
- Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Tensorflow, MobileNetV2
- Infrastructure: Streamlit

**Thank
you!**

